

Loss Given Default Model

A Retail LGD Scorecard with Downturn Adjustment

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1 What this project does

Loss Given Default (LGD) is the fraction of exposure that is lost when a borrower defaults, after all recoveries. It is one of the three components of the expected loss identity used across Basel IRB and IFRS 9:

$$EL = PD \cdot LGD \cdot EAD.$$

This project builds a retail LGD scorecard on a synthetic panel of $\sim 15,000$ defaulted facilities across mortgage, auto, personal secured, and unsecured collateral, spanning ten origination vintages (2019H1–2023H2). The model estimates realised LGD from facility and borrower features, calibrates by collateral bucket, and derives a downturn-adjusted LGD as required by Basel IRB.

2 Data

Each row is one defaulted facility observed to workout resolution. Features cover collateral type, exposure at default, loan-to-value at default, days past due at workout start, workout duration, borrower age and employment status, and vintage-level macro context (unemployment, house-price index growth). The target is realised LGD drawn from a Beta distribution whose mean depends on the features.

Collateral	Count	Mean LGD	Std LGD
Mortgage	6 266	0.224	0.145
Auto	2 729	0.428	0.166
Personal secured	1 842	0.497	0.168
Unsecured	4 163	0.717	0.127

Table 1: Realised LGD by collateral type on the full defaulted-facility population. Mortgage collateral recovers most; unsecured recovers least. Std within each bucket reflects idiosyncratic variation independent of the systematic features.

The macro context per vintage is embedded so downturn cohorts (2020H1–H2 pandemic, 2022H2 inflation shock) show elevated realised LGD.

3 Model

Three models are fit and compared on out-of-time data (train 2019H1–2022H2, test 2023H1–2023H2):

- **Fractional logit** (statsmodels GLM with Binomial family and logit link, quasi-likelihood scale). Handles $[0, 1]$ outcomes without requiring the Beta distribution assumption.
- **OLS on logit-transformed LGD** as a linear baseline. Simple, interpretable, but does not respect the boundary.
- **XGBoost** on logit-transformed LGD as a nonparametric ML comparator, back-transformed for prediction.

Features are standardised (numeric) and dummy-encoded (categorical: collateral type, employment status). LTV at default is imputed to 1.0 for unsecured exposures where the concept does not apply.

4 Results

4.1 Out-of-time model comparison

Model	MAE	RMSE	R^2	Mean pred	Mean true
Fractional logit	0.100	0.126	0.724	0.442	0.439
OLS-logit	0.105	0.136	0.681	0.428	0.439
XGBoost	0.104	0.132	0.696	0.414	0.439

Table 2: OOT metrics on 3,639 defaulted facilities. Fractional logit wins on all three headline metrics. XGBoost narrowly beats OLS on RMSE and R^2 . Mean-predicted is closest to mean-true for fractional logit, indicating no systematic bias at the portfolio level.

Fractional logit is chosen as the production model on the grounds of best OOT performance, interpretable coefficients, and no boundary violations by construction.

4.2 Calibration

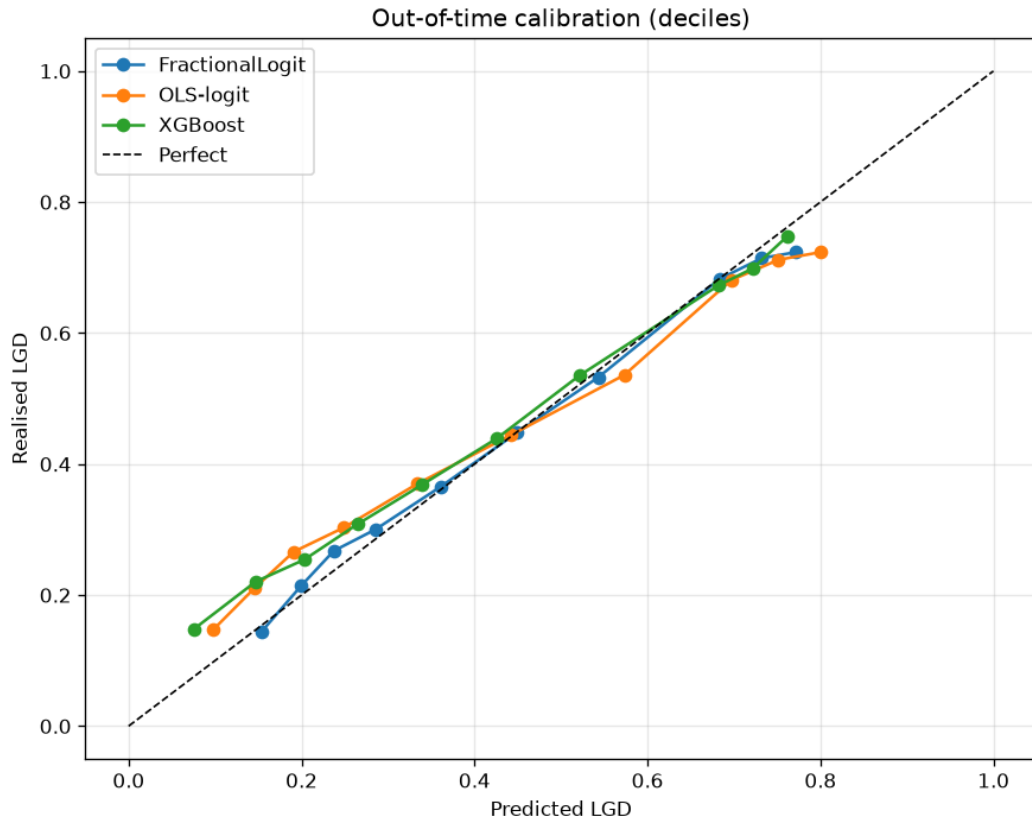


Figure 1: Decile calibration of the three models on the OOT cohort. Fractional logit tracks the diagonal most closely. All three models underestimate at the highest-LGD decile, driven by rare cases with negative recoveries (collection costs exceed collateral).

4.3 LGD by collateral type (OOT)

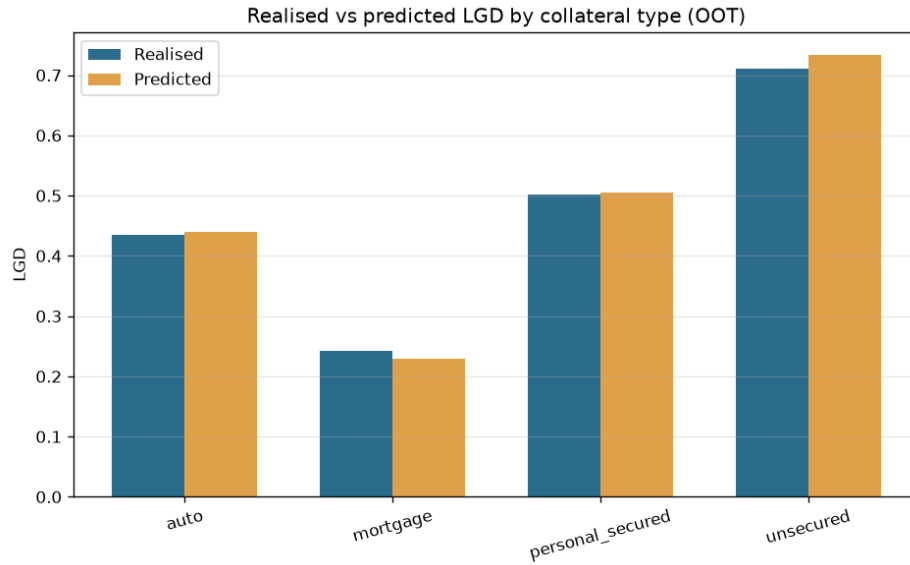


Figure 2: Realised versus predicted LGD by collateral type on the OOT cohort. The model reproduces the collateral-level ordering (unsecured $\gg$ personal secured $>$ auto $>$ mortgage) with predicted means within 3 pp of realised for all four buckets.

4.4 Downturn LGD

Basel IRB requires LGD estimates to reflect stressed conditions, not just the long-run average (LRA). One conservative recipe is to take the 90th percentile of the predicted LGD distribution as the downturn point estimate per bucket. The uplift is the increment over LRA:

Collateral	LRA LGD	Downturn P90	Uplift
Mortgage	0.221	0.343	+0.121
Auto	0.427	0.577	+0.150
Personal secured	0.501	0.679	+0.178
Unsecured	0.712	0.868	+0.156

Table 3: Downturn-LGD uplift per collateral bucket. Personal secured carries the largest absolute uplift because its LTV distribution extends into over-100% territory, which the model penalises heavily.

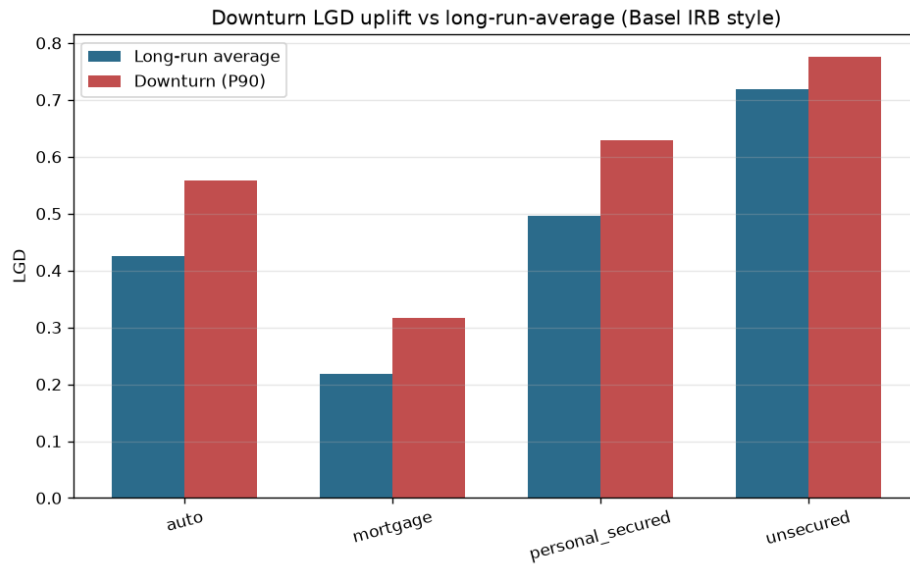


Figure 3: Long-run-average and downturn LGD side-by-side. Downturn values would be used for regulatory capital and IFRS 9 lifetime ECL when the forward-looking macro path is stressed.

4.5 Feature importance

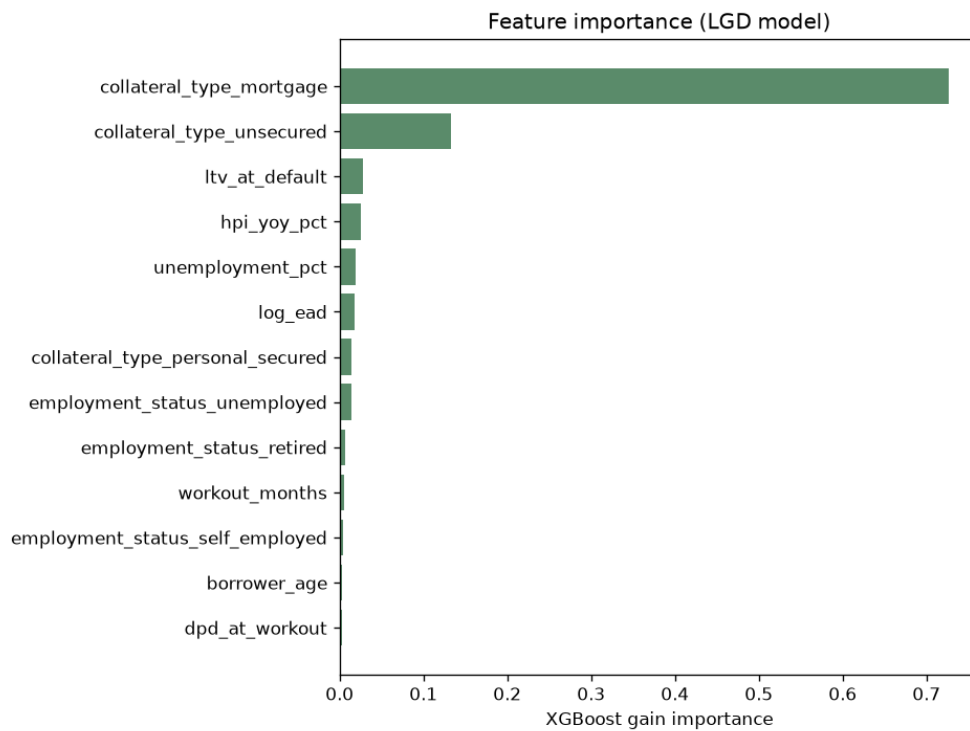


Figure 4: XGBoost gain-based feature importance. Collateral-type dummies carry the most signal (as expected: bucket sets the baseline). Loan-to-value at default and macro variables (HPI, unemployment) are the strongest continuous predictors.

4.6 Realised LGD distribution

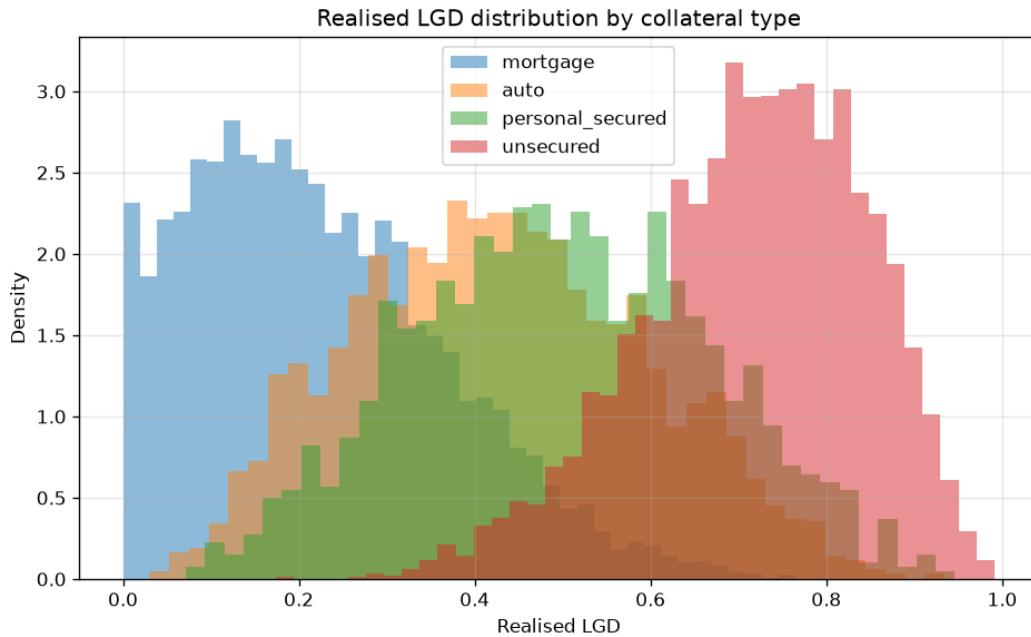


Figure 5: Realised LGD distribution by collateral type. The four components have visibly different centre and spread. Mortgage distribution has a heavy left tail (full recovery); unsecured has a narrow high-LGD peak with almost no full-recovery mass.

4.7 Feature stability (PSI)

Feature	PSI (train vs OOT)
Unemployment rate	17.11
HPI YoY growth	17.63
EAD	0.008
LTV at default	0.003
Days past due	0.003

Table 4: PSI between train and out-of-time periods. Macro variables show very high PSI because the OOT window (2023) has different macro conditions from the train window (2019–2022, which includes the pandemic stress). This is by design and does not indicate model misspecification. Idiosyncratic features are stable.

The macro-level PSI is the informative half of the story here: it tells us that the model must be re-estimated or the macro overlay recalibrated whenever a new macro regime arrives. Idiosyncratic features are stable across time, which supports the use of the scorecard on new-vintage originations.

5 Basel IRB and IFRS 9 fit

- **Basel IRB LGD requirements:** The scorecard produces collateral-level LGD estimates with downturn adjustment via the P90 quantile approach. The long-run-average LGD is computed on the full train sample which spans a full economic cycle (2019–2022), meeting the "sufficient time span" requirement.

- **IFRS 9 lifetime LGD:** The macro-conditional feature set (unemployment, HPI YoY) allows conditioning the LGD estimate on forward-looking macroeconomic scenarios, which is how lifetime ECL is generated in stage-2 and stage-3 classifications.
- **Not fully compliant on:** single-vintage bucket granularity may be insufficient for capital calculations that require finer risk-driver segmentation; a production model would decompose by product family, geography, and origination channel.

6 Limitations

- Synthetic data has clean, non-correlated features. Real recovery data has heavy dependence on macro cycles, jurisdiction, and collateral illiquidity that are only sketched here.
- Workout duration is modelled as observed, but in production many defaults are still in workout at reporting date and require survival modelling for time-to-resolution.
- Collateral values at default are treated as observed. In reality collateral is re-appraised at workout, and appraisal uncertainty is a first-order source of LGD variance.
- Downturn P90 is a simple rule. A production model would use a macro-conditional simulation to derive the downturn LGD.

7 Files

- `generate_data.py` — synthetic default panel generator
- `train.py` — three models, calibration, downturn, PSI
- `outputs/` — metrics, PSI, downturn LGD as CSV
- `figures/` — five diagnostic plots